Neural Decoder for Hand Trajectory Estimation

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Abstract— **This work addresses the problem of neural decoding of hand trajectories from population spike train data. We propose a causal decoding framework that estimates the two-dimensional hand position from neural activity recorded during reaching movements. The neural signals are first transformed into causal features using binned spike counts over a short temporal history window. To reduce dimensionality and noise, principal component analysis (PCA) is applied to the feature space. A ridge regression model is then trained to map the reduced features to hand positions. Additionally, a direction-based strategy is employed, where separate decoders are trained for each movement direction and selected during inference based on neural activity patterns. The proposed method achieves a root mean squared error (RMSE) of 14.35 cm on the validation data, demonstrating stable and reliable decoding performance.**

Keywords—Neural decoding, Brain-machine interface, PCA, Ridge regression, Trajectory estimation

I. INTRODUCTION

Brain–machine interfaces (BMIs) aim to translate neural activity into control signals for external devices, with important applications in neuroprosthetics and assistive technologies. A core problem in this area is neural decoding, where recorded neural signals must be used to estimate intended movement.

The present task considers decoding the two-dimensional hand trajectory of a monkey during a reaching experiment. The dataset contains spike train recordings from 98 neural units collected while the monkey repeatedly reached towards targets in eight different directions. For each trial, neural activity is recorded at 1 ms resolution together with the corresponding hand position trajectory, and the objective is to estimate the hand position in the horizontal plane over time from neural activity alone. Since the competition requires causal decoding, only past and present neural information can be used at each time point.

This problem is challenging for several reasons. First, neural spike trains are sparse and noisy. Second, the data are high-dimensional, as activity from many neural units must be integrated over time. Third, the relationship between neural activity and movement is dynamic, meaning that both movement direction and fine-grained trajectory information must be captured by the decoder.

A simple baseline pipeline was first implemented using direction templates for coarse classification, followed by principal component analysis (PCA) and ridge regression for trajectory prediction. This baseline was computationally efficient, easy to interpret, and suitable as an initial reference method. An improved competition pipeline was later developed, in which direction classification was refined using Linear Discriminant Analysis (LDA) and trajectory prediction was performed using Principal Component Regression (PCR). Although both methods are based on relatively lightweight linear models, the latter provides a more structured treatment of classification and regression and achieved stronger decoding performance.

Comparison between the baseline pipeline and the improved competition pipeline provides insight into how different design choices affect decoding accuracy, robustness, and interpretability.

II. METHOD

Two pipelines are explained in this section, including original version "**Per-direction, Per-time PCA + Ridge Decoder"** and improved one.

B. Per-direction, Per-time PCA + Ridge Decoder

The second pipeline used a causal PCA-ridge decoder with separate models for movement direction and decoding time. Spike trains were binned in 20 ms windows, the latest K = 5 bins were concatenated into a causal history feature, and all aligned training samples were pooled to learn a single global PCA projection. Ridge regressors were then trained in the reduced space to estimate the two-dimensional hand position.

For a decoding time t, spike counts from the latest K = 5 bins (bin size 20 ms) were concatenated to form a causal feature vector:

$$\varphi(t) = [c_{t-K+1}^T, c_{t-K+2}^T, \ldots, c_t^T]^T \qquad (1)$$

All aligned training features from all directions and time points were pooled to compute a single PCA projection:

$$z(t) = U^T (\varphi(t) - \mu) \qquad (2)$$

A separate weight matrix was learned for each direction d and decoding time t_i. At test time, cumulative spike counts up to t_{start} = 320 ms were first matched to direction templates by nearest distance. The latest available time

index was then selected, the current feature was projected by the stored PCA basis, and the corresponding model produced the position estimate:

$$\hat{p}(t) = W(d,i)\ [\ z(t)^T,\ 1\ ]^T \qquad (3)$$

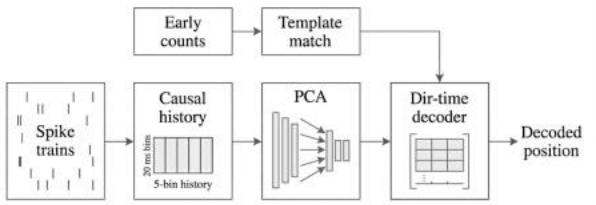

Fig. 1. Simplified overview of the per-direction, per-time PCA + ridge decoder.

B. Improved Pipeline: This pipeline includes four stages. First is preprocessing, where discrete spikes from the training data are converted into continuous numerical inputs (**Binning**) by averaging the firing rate over a given period (Δt) to smooth the data, provide a richer gradient for pattern discovery, and help eliminate jitter in each spike while preserving the overall changes across spikes related to the arm's movement. The equation is shown below:

$$V_i = \frac{1}{\Delta t}\sum_{t=t_1}^{t_2} S_i(t)$$

Where $S_i(t)$ is the binary spike data (0 or 1) at time t, and Δt is the duration of the bin $(t_2 - t_1)$.

Second is dimensionality reduction. Principal Component Analysis (**PCA**) is used to extract the most informative features from the bins, while discarding the remaining data to reduce processing time. A fixed number of feature dimensions is used to ensure the same number of components are passed to the Linear Discriminant Analysis (LDA) for classification. The equation of PCA is shown below:

$$Z_{pca} = W_{pca}^T(f - \mu)$$

Where W_{pca}^T is the matrix of the top fixed number of eigenvectors of the covariance matrix and μ is the mean feature vector.

To reduce root mean squared error (RMSE), the firing rate bins are classified into 8 directions based on their highest firing rates. Each direction has a specific weighted method for prediction instead of using a single general method for all 8 directions. Here, **LDA** and Nearest Centroid (**NC**) methods are used. During LDA, the PCA components are projected into a certain dimensional space to maximise the spacing between the data bins, forming 8 clusters (directions). The maximum ratio of between-class scatter () to within-class scatter () is calculated based on the following equation for maximising the spacing between clusters:

$$J(W) = \frac{|W^T S_B W|}{|W^T S_W W|}$$

Third is direction classification, the mean value of each separated cluster (direction) in LDA is found as the centre (centroid). NC is used to measure the distance between the centres and the bin data and classify it as the direction whose centre is closest. The equation of NC is shown below:

$$\hat{C} = arg\ \min_k \sum_{j=1}^{M_{LDA}} (Z_j - \bar{Z}_{kj})^2$$

Where Z is the current LDA-projected vector, $\bar{Z}_k$ is the mean vector for direction k, and $\hat{C}$ is the predicted direction.

Fourth is position estimation, where the spike trains are split into smaller bins to track the hand's path with higher temporal resolution. Also, cumulative bins are used for more stable and precise prediction. Each direction has its linear regression Principal Component Regression (**PCR**) model, with specific weights to filter out individual-neuron noise and multicollinearity. The feature space of each direction's data is transformed into N-1 (N: number of trials) principal components (PCs) to prevent singularities, decorrelate the features for calculations, and prevent overfitting. The raw features, with the mean subtracted, are first projected onto the eigenvector matrix to create the PCs, which are then mapped by the Regression weights to calculate the predicted coordinates. The equations are as shown below:

$$x(t) = f_{reg}^T \beta_x + \alpha_x;\ \ y(t) = f_{reg}^T \beta_y + \alpha_y;$$

Where f_{reg} is the vector of neural firing rates, β is the vector of regression coefficients, and α is the bias.

The high-level flow of this pipeline is shown in Fig. 2.

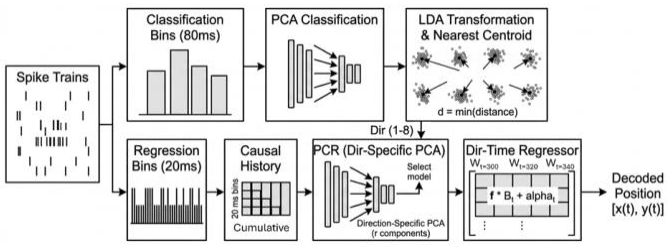

Fig. 2. High-level flow of improved pipeline.

III. Results

Evaluation was performed using the provided competition harness. The 100-trial dataset was randomly partitioned — using a fixed seed for reproducibility — into 50 training trials and 50 test trials, across all 8 reaching directions. The model was trained once on the training set, then evaluated causally on the test set: for each test trial, the decoder was called iteratively

from t = 320 ms to the end of the trial in 20 ms steps, with only spikes up to the current time available at each step. Performance was measured as the root mean squared error (RMSE) of the decoded versus actual two-dimensional hand position, pooled across all time steps, trials, and directions as shown in TABLE I.

TABLE I. OVERALL MEAN RMSE ACROSS ALL 8 DIRECTIONS

Model	RMSE overall	RMSE_x	RMSE_y	R^2_x	R^2_y
Baseline	13.267	8.542	15.125	0.98	0.91
Improved	8.679	4.972	5.926	0.99	0.99
Error reduction %	-35%	-42%	-61%		

Per-direction RMSE was also computed to assess directional decoding consistency as shown in Figure 1. The baseline model shows a much higher magnitude of error in direction numbers 3, 4, and 5 as compared to the LDA-PCR pipeline.

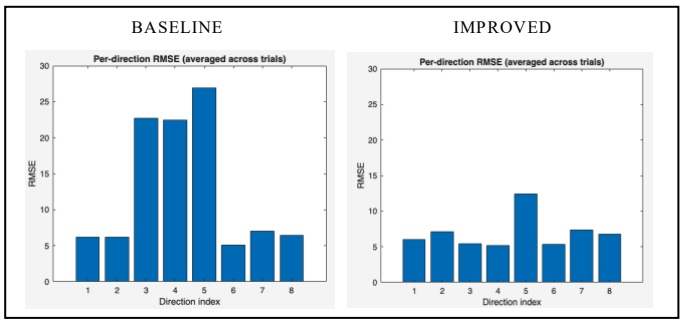

Figure 1. RMSE per direction for both pipelines. (average)

From Figure 2. the baseline model shows large MSE spikes for a small subset of trials, suggesting sensitivity to outlier trials rather than generalised overfitting.

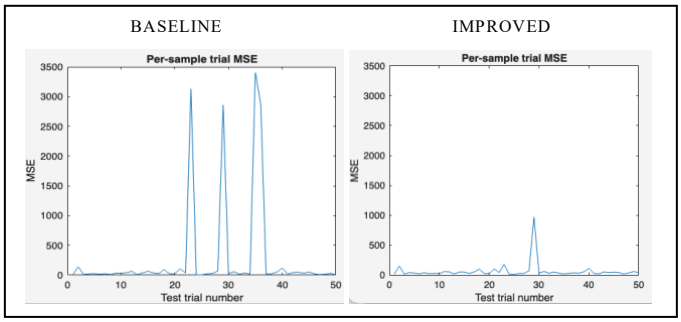

Figure 2. MSE per individual test trial for both pipelines.

The residuals histograms for X and Y directions comparing both pipelines are shown in Figure 3. The x-residual distribution for the baseline model shows peakedness around 0, but has a wider spread or deviation. This indicates the presence of outliers causing distortions in prediction. This is observed much more keenly in the y-direction, which has a bigger contribution to the high RMSE of the base pipeline.

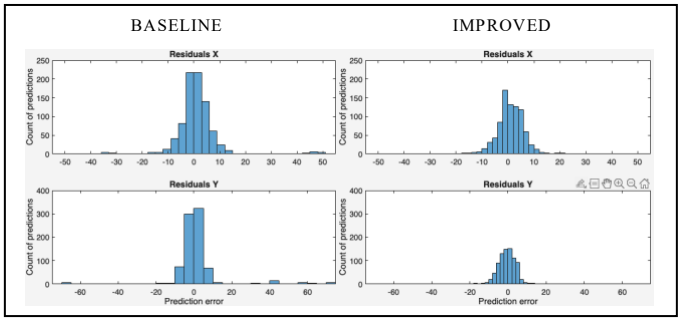

Figure 3. Residual histogram for X and Y directions for both pipelines.

Overall, the proposed method of the winning team demonstrated a much more stable and reliable decoding performance for a 35% lower RMSE compared to our initial model.

IV. DISCUSSION

The PCA + LDA + PCR pipeline proved to be much more accurate than the Per-direction, Per-time PCA + Ridge Decoder. The principal reason is that a single, well-estimated global PCA basis (pooled across many samples) produces stable low-dimensional features for PCR, while LDA (or template matching) on early firing rates captures the dominant direction signal reliably.

A. Effect of Direction Classification Method

Our initial cosine/correlation-based template matching method was simple but tends to conflate noise with direction signal. The PCA-LDA with nearest centroid method in the improved pipeline avoided this by maximising between-class separation.

B. Effect of Regression Method

Our single global PCA based ridge regression penalises large weights, which is good for noisy/collinear features. In contrast, the PCR with OLS in the improved pipeline uncouples neurons via PCA, uses N−1 components to avoid singularity, hence does not require explicit regularisation penalty.

Both methods have their trade-offs: ridge regression is more robust to overfitting via λ, while PCR relies on component truncation for regularisation. The difference lies in the role of PCA in the pipeline.

C. Direction-Specific Performance Analysis

The direction #5 in the trial was relatively harder to decode for both algorithms (see Figure 1.), but for the baseline method direction #3 and #4 caused a lot of error as well. In addition, the y-direction was harder to train in the base pipeline. Investigating the outliers in the tuning curves at the preprocessing stage, could potentially shed some light on these anomalies. Actual trajectores in the x and y direction are shown for direction #5 in Figure 4. where the y trajectory shows more variability.

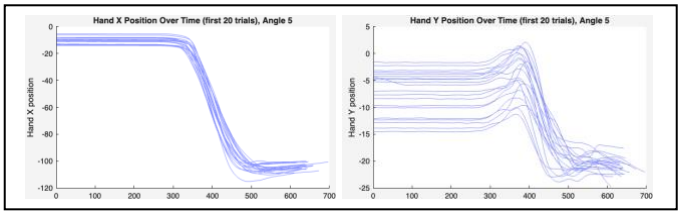

Figure 4. Sample trials for direction #5 time series (Position X,Y)

D. *Robustness to outlier trials*

The improved model's robustness to outlier trials stems from four structural differences. First, its growing feature vector adapts naturally to longer trials, unlike the base model. For the latter, the classification update is fixed at training. the decoder only ever sees the most recent 220 ms — it discards much of the trial history. Second, direction is re-estimated at four time checkpoints (320–560 ms), allowing recovery from early misclassification. Third, LDA provides a more discriminative classification boundary than template correlation. Fourth, PCR with N−1 components decorrelates the neural features, preventing individual outlier neurons from dominating the regression. Together, these factors explain why the MSE spikes visible in the baseline (Figure 2) are largely absent in the improved model.

Even so, for the LDA + PCR method, T_end is hard-coded 560 ms. These may not generalise to longer trials and may limit decoding which operates until 100 ms before the end of trial. A small percentage of trials lie well beyond this cutoff as shown in the histogram of **Error! Reference source not found.**

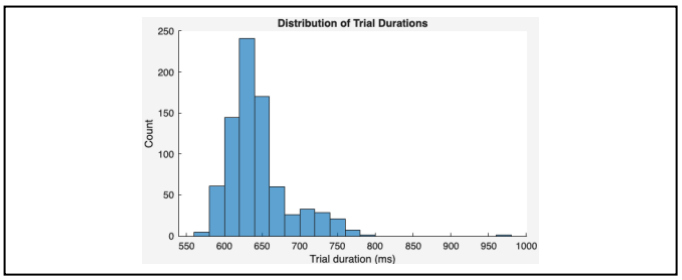

Figure 5. Histogram of trials by time duration

E. *Limitations*

- No non-linear methods explored (e.g., Kalman filter, neural networks)
- Single train/test split — no cross-validation reported

Restricted runtime requirements and development complexities made these non-viable. Future work to improve decoding performace would entail these components.

References

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